Answers

Worksheet

# Equivalent multiplicative relationships

## Task A: Equivalent ratios

1) Tick which ratio tables are equivalent to the statement:

"For every 3 ticks, there are 4 crosses."

| Ticks | Crosses |
|---|---|
| 3 | 4 |

✓

| Ticks | Crosses |
|---|---|
| 6 | 9 |

$\times\frac{1}{2}$

| Ticks | Crosses |
|---|---|
| 1.5 | 2 |

✓

×5

| Ticks | Crosses |
|---|---|
| 15 | 20 |

✓

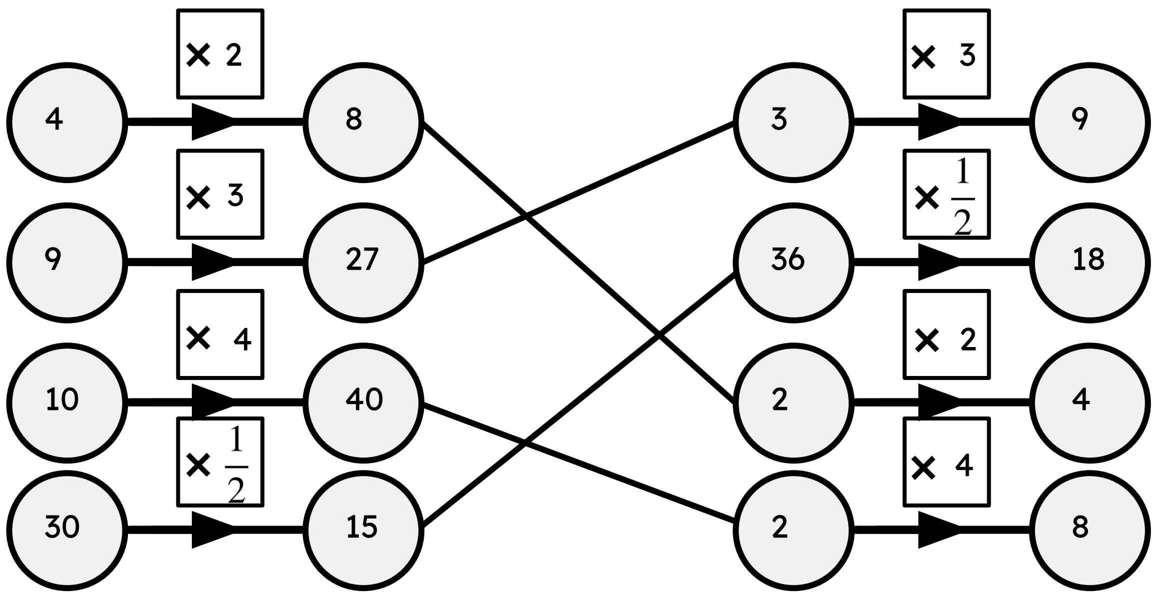
×3

| Ticks | Crosses |
|---|---|
| 9 | 12 |

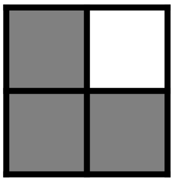
✓

| Ticks | Crosses |
|---|---|
| 4 | 5 |

2) Pair the equivalent function machines, left to right, and identify the multiplier.

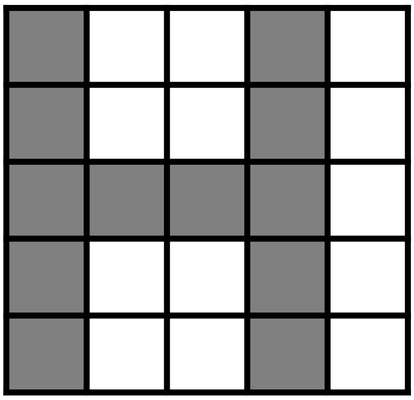


## Task A: Equivalent ratios



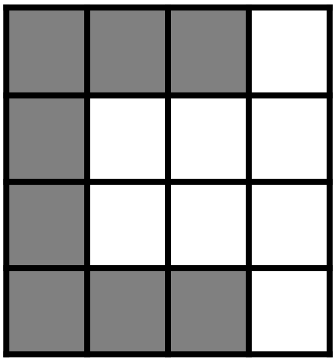
3a) Shade in to make the letter "L" using the ratio:

3 shaded to 1 unshaded.



b) Shade in to make the letter "H" using the ratio:

12 shaded to 13 unshaded.



c) Shade in to make the letter "C" using the ratio:

1 shaded to 1 unshaded.

4) Identify the multipliers and missing numbers for the following function machines.

a) $1 \xrightarrow{\times 5} 5$

b) $9 \xrightarrow{\times \frac{10}{9}} 10$

c) $15 \xrightarrow{\times \frac{2}{3}} 10$

d) $50 \xrightarrow{\times \frac{3}{5}} 30$

## Task B: Using equivalent ratios

1) Using the ratio table, fill in the blanks.

a)

| Stars | Hearts | Total |
|---|---|---|
| 2 | 3 | 5 |
| 20 | 30 | 50 |

For every 2 stars, there are 3 hearts.

$\frac{2}{3}$ is the fraction of stars.

When there are 50 shapes, there are 20 stars and 30 hearts.

b)

| Boys | Girls | Total |
|---|---|---|
| 6 | 5 | 11 |
| 48 | 40 | 88 |

For every 6 boys, there are 5 girls.

$\frac{5}{11}$ is the fraction of girls.

When there are 88 pupils, there are 48 boys and 40 girls.

2) To make screed, the ratio of sand to cement is for every 1 kg of sand, use 4 kg of cement.

a) Show this in a ratio table.

| Sand | Cement | Total |
|---|---|---|
| 1 | 4 | 5 |
| 3 | 12 | 15 |

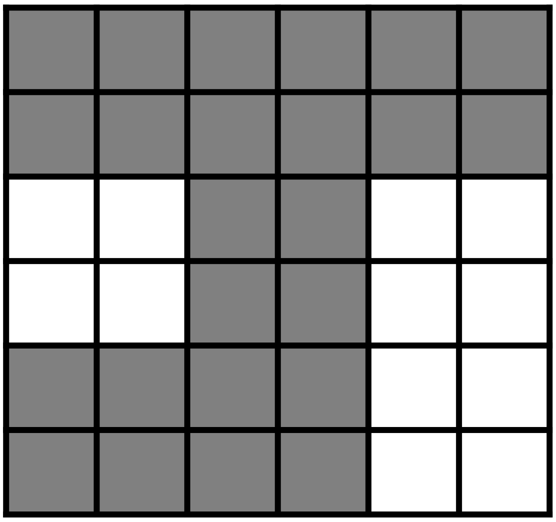
b) Andeep needs 15kg in total. How many kg of sand will he need?

*He would need 3 kg of sand.*

c) What fraction of the screed is sand? $\frac{1}{5}$

## Task B: Using equivalent ratios

3) Jacob spilt ink all over the maths question and his working out, but he correctly shaded in the grid. Work out the maths questions and working out.



For every 2 shaded there are 1 unshaded

The fraction that is shaded is $\frac{2}{3}$

Using 36 squares, $\frac{2}{3} \times \frac{12}{12} = \frac{24}{36}$

I need to shade in 24 squares.

4) The ratio of the three different panels of fencing are:

Panel length A × 1.5 = panel length B

Panel length A × 2 = panel length C

The total length of the fencing is 36 cm. Work out the length of each panel.

| A | B | C | Perimeter |
|---|---|---|---|
| 2 | 3 | 4 | 9 |
| 8 | 12 | 16 | 36 |

*A has length 8 cm, B has length 12 cm and C has length 16 cm.*